

Ruixin Guo

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Research interests

Statistical and *analytical* aspects of theoretical machine learning, particularly statistical learning theory.

Applied machine learning: deep learning, recommender systems, large language models.

High-performance computing: code parallelization and algorithm optimization.

Education

Kent State University, PhD Candidate in Computer Science, 2021 – .

Advisor: Ruoming Jin

China University of Geosciences (Wuhan), MSc in Computer Science, 2018 – 2021.

Advisor: Feng Zhang

China University of Geosciences (Wuhan), BS in Information Security, 2014 – 2018.

Work Experience

Research Scientist Internship, OppyAI Inc., May 2026 – Dec 2026.

Advisor: NhatHai Phan

Focusing on secure and privacy-preserving LLMs.

Honors and scholarships

NeurIPS Financial Aid Award, 2025.

China National Scholarship, China University of Geosciences (Wuhan), 2019.

Silver Award in the 6th Parallel Application Challenge, China Computer Federation (CCF), 2018.

Publications

Journals

Label-Free quantification of protein density in living cells

RJ Clements, R Guo, JC Petrucci, MA Model. *Current Protocols*, 2025.

Measurement of protein concentration in bacteria and small organelles under a light transmission microscope

MA Model, R Guo, K Fasina, R Jin, RJ Clements, LG Leff. *Journal of Molecular Recognition*, 2024.

Earth Observation Data Provenance: A Blockchain-Based Solution

Feng Zhang, Zihao Wang, Ruixin Guo, Guangzhi Qu. *IEEE Transactions on Industrial Informatics*, 2024.

DS-ADMM++: A novel distributed quantized ADMM to speed up differentially private matrix factorization

Feng Zhang, Erkang Xue, Ruixin Guo, Guangzhi Qu, Gansen Zhao, Albert Y Zomaya. *IEEE Transactions on*

Industrial Informatics, 2021.

BaPa: A novel approach of improving load balance in parallel matrix factorization for recommender systems

Ruixin Guo, Feng Zhang, Lizhe Wang, Wusheng Zhang, Xinya Lei, Rajiv Ranjan, Albert Y Zomaya. IEEE Transactions on Computers, 2020.

Optimizing the confidence bound of count-min sketches to estimate the streaming big data query results more precisely

Ruixin Guo, Erkang Xue, Feng Zhang, Gansen Zhao, Guangzhi Qu. Computing, 2020.

Optimizing FHEW with heterogeneous high-performance computing

Xinya Lei, Ruixin Guo, Feng Zhang, Lizhe Wang, Rui Xu, Guangzhi Qu. IEEE Transactions on Industrial Informatics, 2019.

Conferences

Generalizing Linear Autoencoder Recommenders with Decoupled Expected Quadratic Loss

Ruixin Guo*, Xinyu Li*, Yang Zhou, Hao Zhou, Ruoming Jin. International Conference on Learning Representations (ICLR), 2026.

PAC-Bayes bounds for Multivariate Linear Regression and Linear Autoencoders

Ruixin Guo, Ruoming Jin, Xinyu Li, Yang Zhou. Neural Information Processing Systems (NeurIPS), 2025.

Under Review

Wrapped Gaussian Mixture on the Low-Rank Matrix Manifold

Ruixin Guo, Ruoming Jin. Submitted to ICLR 2027.

APlaud: Adaptive Personalized Low-Rank Decomposition for User-Specific LLM

Ruoming Jin, Xinyu Li, Jianfeng Zhu, Ruixin Guo, Zhi Liu. ArXiv:2609.04738.

Geometry-Aware Personalized Preference Optimization via Gradient Alignment

Ruoming Jin, Xinyu Li, Hao Zhou, Jianfeng Zhu, Ruixin Guo, Feodor Dragan, Lei Xu, Haixun Wang, Yang Zhou.

PLUME: Personalization of Large Language Models through Low-Rank User Modulation and Shared Subspaces

Xinyu Li, Hao Zhou, Jianfeng Zhu, Julina Maharjan, Ruixin Guo, Feodor Dragan, Ruoming Jin. ArXiv:2609.04715.

Academic Service

Reviewer for NeurIPS 2026 and ICLR 2027.

Teaching Assistant Experience

I served as a *lab assistant* or a *grader* for these courses:

Machine Learning and Deep Learning, CS 54202, Spring 2024.

Design and Analysis of Algorithms, CS 46101, Fall 2022.

Operating Systems, CS 33211, Fall 2022.

Discrete Structures for Computer Science, CS 23022, Spring 2026, Spring/Fall 2025.

Computer Science IA: Procedural Programming, CS 13011, Fall 2024.

Programming for Problem Solving in Sciences, CS 10062, Fall 2025/2024/2023.

Computer Science Principles, CS 10051, Spring 2023.